AACES 2050: Electra.aero Subcontractors

Internal Memorandum

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SUBJECT: Outline of Environmental Impact Modeling with the LAE and U-M

The University of Michigan teams' understanding of the integration of (non-acoustic) environmental impacts into the FINCH tool is summarized by six items:

1. Production of exhaust species such as CO, CO_2, H_2O, SO_x, and NO_x has a substantial effect on the climate impact of a flight. This list is comprehensive and additional species such as unburned hydrocarbons and particulate matter will not be considered. FINCH needs to predict the production of exhaust species during flight, so that their subsequent climate impact can be quantified by a metric including global warming potential (GWP), effective radiative forcing (ERF), and equivalent CO_2 emission (CO_2e). The climate impact metric may then be used as an optimization objective or as a constraint in the minimization of another metric, such as gross aircraft weight.

   Internal to FINCH, a climate impact function will require the following inputs, each reported as a $1 \times n$ matrix of values at each time index in the mission, where n is the number of time steps in the mission:

   Altitude (h or z): Aircraft altitude.

   Ambient State (T_a, P_a, $M_a = 0$): Full thermodynamic state of the ambient air.

   Exhaust Gas Temperature: Temperature of the gas exiting the core and bypass nozzles.

   Fuel-to-Air Ratio (f): Ratio of the fuel mass flow rate to the air mass flow rate.

   Fuel Type: Qualitative value, e.g. Jet A, LH_2, gaseous hydrogen.

   Fuel Flow Rate ($\dot{m}_f$): Fuel mass flow rate.

   Full Combustor State ($T_{t,4}$, $P_{t,4}$, T_4, P_4): Full thermodynamic state of the air/fuel inside the combustion chamber.

   Jet Velocity (v_e): Speed of exhaust gases exiting the core and bypass nozzles.

   Time Index: Elapsed mission time at each node.

The inputs as listed above are fully sufficient to predict the production of CO, CO_2, H_2O, SO_x, and NO_x, and their subsequent effects on climate impact using a metric such as GWP, ERF, CO_2e, or a more appropriate alternative at the discretion of the LAE. This metric should be reported at each time step such that time integration is performed on the Aviary level, not on this function's level.

2. Contrails, when formed, have a substantial contribution to the climate impact of a flight. Therefore, it is necessary to build functionality into the FINCH tool to accomplish two objectives:

   (a) During on-design (sizing) analysis, modify aircraft to include a contrail avoidance ability. This ability is added to the aircraft by modifying altitudes in the design mission profile, sizing the aircraft to fly a more demanding mission. The aircraft, which would normally be over-sized, now has the capability to avoid contrails using altitude control while cruising in off-design missions. This necessitates the creation of a mission profile modification function, described by item 3 below.

   (b) During off-design (operational) analysis, the aircraft uses its ability to avoid contrails by informing the operation with desired cruise altitudes. This necessitates the use of both a contrail climate impact function and atmosphere model, which are described by items 4 and 5 respectively.

3. As described in item 2.a, a function to modify a mission profile is required. The function leverages LAE's expertise in contrails, and recommends a mission profile which represents the "worst case" scenario (i.e., the most energy-demanding mission profile that sizes the aircraft for contrail avoidance). A standard mission profile is input, parametrized by altitude, Mach number, and design range. The function then outputs a profile with modified altitudes and Mach numbers, although design range remains unchanged. This approach is a pre-optimization computation, and avoids in-the-loop interaction between contrail formation and aircraft sizing. If the mission profile (trajectory) is to be changed *during* the sizing process, it should immediately be brought to U-M's attention.

4. As described in item 2.b, a function to predict climate impact of contrails is required. This function takes advantage of the inputs described in item 1, as well as any additional inputs required by an atmosphere model described in item 5. The function then outputs the same climate impact metric down-selected for item 1 (e.g. ERF).

5. Contrails cannot form under a standard atmosphere model. Therefore, it is necessary to construct a family of atmosphere models which allow for the formation of contrails. Aircraft operations will be simulated using this family of models, and the effect of contrail formation on climate impact quantified, using the metrics mentioned in item 1 above. Each model returns atmospheric conditions necessary to predict contrails' climate impact. These conditions include ambient temperatures, pressures, and anything else required by item 4 (e.g. humidity, if required) as functions of altitude.

6. All functions written for FINCH integration will include derivatives of every one of that function's outputs with respect to every one of that function's inputs. Functions are self contained and do not need to trace derivatives outside of their own input/output spaces. Exception is that FINCH does not require derivatives of item 3 because this is called outside of the optimization loop, assuming that we do not optimize mission profile during the sizing process.

LAE will build **four functions**, corresponding to items 1, 3, 4, and 5. The flowcharts below summarize the functions described in each item.

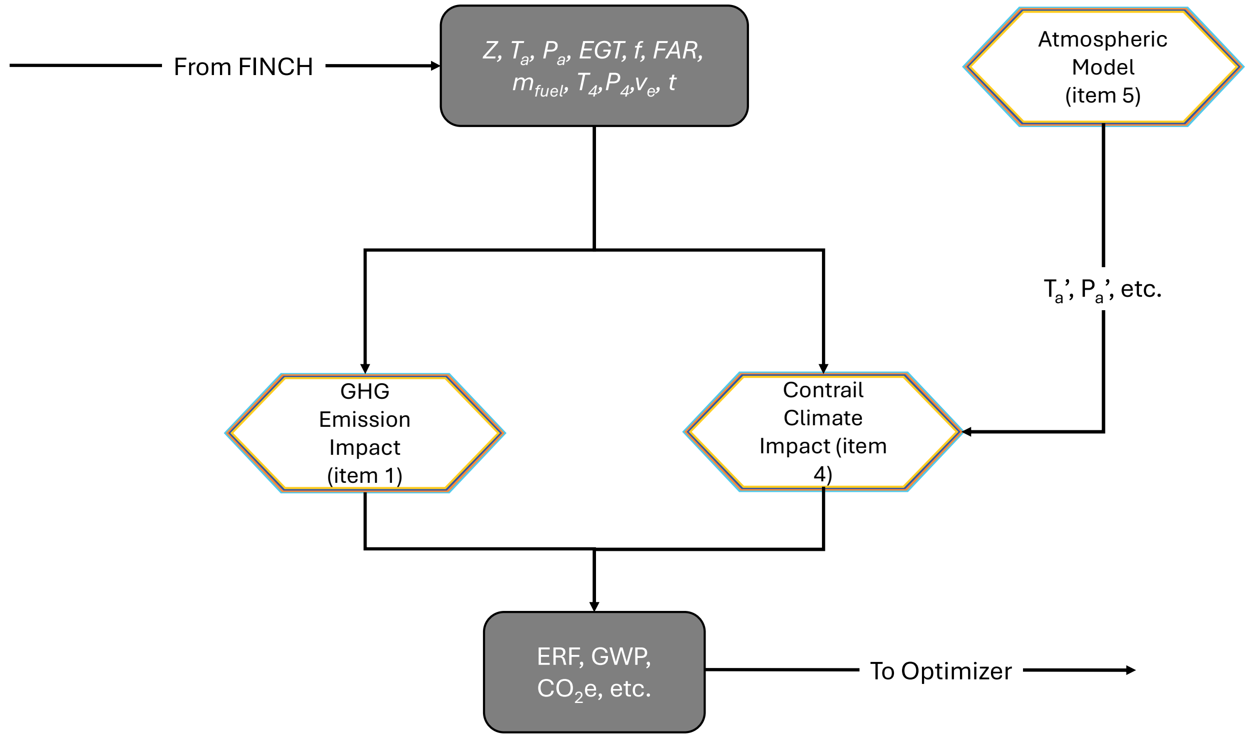

Figure 1: Items 1, 4, and 5 placement in operational analysis. Gray boxes represent interaction with FINCH outside of LAE contributions.

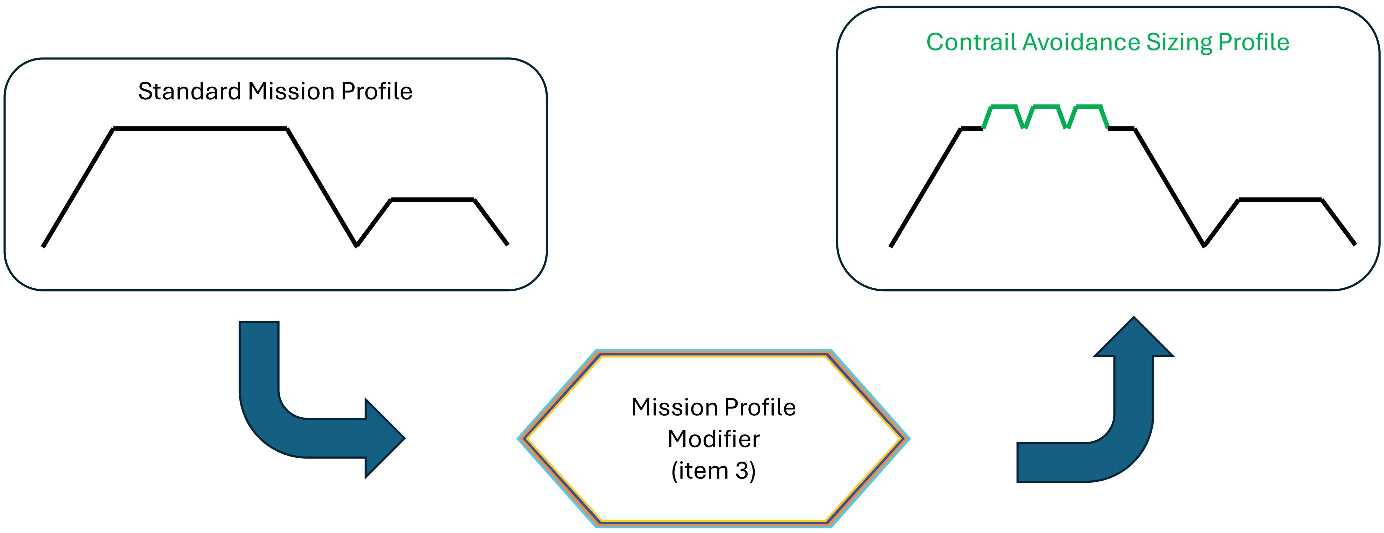

Figure 2: Mission profile modifier, created by LAE, to inform aircraft sizing of representative demands on the aircraft when avoiding contrails.